JACOB RICKEY

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SUMMARY

Master of Information and Telecommunication Systems student with hands-on experience in network automation, cloud infrastructure, and cybersecurity. Combines graduate coursework in 5G/LTE architecture and penetration testing with practical project work spanning Python, Azure, and emulated network environments.

EDUCATION

Master of Information and Telecommunication Systems, Ohio University *Expected Dec 2026*

Bachelor of Science in Information and Telecommunication Systems, Ohio University *May 2025*

TECHNICAL SKILLS

Networking: TCP/IP, OSPF, BGP, IPsec/VPN, VyOS, GNS3, Wireshark, 5G/LTE

Cloud & Infrastructure: Azure, OpenTofu (Terraform), Ansible, Linux

Security: Penetration testing methodology, vulnerability assessment, regulatory compliance frameworks

Programming & Data: Python (Pandas, SciPy, Matplotlib), SQL, Bash, Java, C++, Anthropic API / MCP

EXPERIENCE

Support Engineering Intern, JBT Marel (Automated Systems Division) *Summer 2025*

- Initialized and deployed 45 next-generation Self-Guided Vehicles (SGVs) from crate receipt through production handoff at a large medical center.
- Developed a radio configuration template for SGVs to maintain reliable Wi-Fi roaming in a complex hospital RF environment.
- Performed preventative maintenance and major repairs including motor swaps and wiring rework on production autonomous platforms.
- Diagnosed safety-system faults spanning sensors, navigation, and control software, applying full-stack troubleshooting from hardware up.

PROJECTS

Natural Language Network Automation via MCP *Fall 2025*

- Built a CLI tool that uses Anthropic's Claude API to translate natural language intent into executable network commands.
- Developed a Python-based MCP server exposing the GNS3 API as LLM-callable tools, enabling automated project and node management on emulated infrastructure.
- Implemented a real-time orchestration pipeline where prompts are reasoned by the model, converted into tool-use blocks, and executed as HTTP requests against the testbed.

Post-Quantum Metadata Leakage Analysis *Fall 2025*

- Simulated encrypted VPN tunnels with StrongSwan in a GNS3 testbed to model links secured by post-quantum cryptography.
- Co-developed Python scripts to capture side-channel metadata and analyze trends across packet size and inter-arrival time.
- Demonstrated that traffic types (industrial control vs. file transfer) remain inferable from metadata despite post-quantum encryption.

CERTIFICATIONS

Wireless Infrastructure Association (WIA) 5G Readiness Certificate *May 2025*

- Covered 4G LTE / 5G architecture, edge computing, and broadband policy. Hands-on lab work with spectrum analyzers and passive intermodulation (PIM) test equipment.